<u>Major task – Fall 2024</u>

It is required to design a rocket to deliver a communication satellite that weighs 10 tons to outer space at an altitude of 300 km from the earth surface. The target time required to achieve this mission is around 9 minutes. The rocket uses liquid hydrogen as a fuel and liquid oxygen as an oxidizer to perform the combustion. You are required to:

a- Show a complete rocket design, including the rocket motor, size, and weight.
b- Show the variation of the dynamic parameters like combustion pressure, temperature, acceleration, thrust, drag, and weight vs time.
c- Build a physical or numerical model using dynamic similarity to show the design concept. ***<u>NO COMBUSTION IS PERMITTED</u>*** rather you can use compressed air. ***<u>(5 points optional)</u>***

Consider isentropic flow through the rocket motor (convergent-divergent nozzle), and consider the following:

- Group submission is allowed, and the maximum number of students in the group is ten.
- The role of each group member must be incorporated in the document at the very beginning of the final report under "Role statement" on a separate page.
- The final report must be submitted through the LMS. The submission file should be uploaded by the team leader.
- Source references must be listed properly in a separate section "References" at the end of the report. Citation of the sources used within the text must be clearly identified. You can use a free citation software like "Mendeley" https://www.mendeley.com/download-reference-manager/
- The submission file size is limited to 20 MB.
- The report will get a plagiarism check. Any report that shows a similarity above 25% will affect the report score. Reports will be checked against open-source databases and all internal reports submitted through the LMS.
- Each student must include and sign the following Plagiarism statement after the cover page:

"I certify that this report is original and has not been submitted for the assessment of another course. I certify that I have not copied in part or whole or otherwise plagiarized the work of other students or persons."

Grading Criteria (Rubric):

	Score
Cover page & Plagiarism statement & Role statement	**5%**
Introduction	
• Effectively presents the objectives and purpose of the project • Describe and display the theory behind the project being conducted (includes any important equations) • Forms an outline for the report	**5%**
Analysis and Design	
• List in paragraph format all tools used • Procedure for the analysis in paragraph form (detailed enough for reproduction) • Include schematics/screenshots of the steps if any • Complete design along with the drawing	**30%**
Result and Discussion	
• Opens with an effective statement of the overall findings • Presents figures and tables clearly and accurately • Presents verbal findings clearly and with sufficient support • Discuss the results found	**30%**
Conclusion	
• Convincingly describes what has been observed • Provides sufficient and logical explanation for the statement • Sufficiently addresses other issues	**10%**
References & Appendix	
• Include citations for all references used • Includes all data presented in the paper (Mathcad, excel, calculations, COMSOL) and equations used • All data is logically organized and properly labeled	**5%**
Overall	
• Overall organization, professionalism, grammar, and proper formatting	**5%**
Presentation	**10%**
Total	**100%**